



AD Hoc Insights

Customer Goods Domain

AtliQ Hardware Codebasics SQL Resume Project Challenge

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Tools Used	MySQL, Power BI
Database	gdb023 (AtliQ Hardware)
Fiscal Year	September 1 st – August 31 st
Data Range	FY 2020 – FY 2021
Task	10 Ad Hoc Business Requests

Executive Summary

This document presents the business insights derived from 10 ad hoc SQL queries on AtliQ Hardware's gdb023 database. Each section summarises the business question, the key data finding, the analytical insight, and a practical recommendation written for a non-technical audience.

S. No	Business Requests
01.	AtliQ Exclusive markets in APAC
02.	Unique product growth FY2020 vs FY2021
03.	Unique product count by segment
04.	Segment with most product growth
05.	Highest & lowest manufacturing cost
06.	Top 5 discount customers in India FY2021
07.	Monthly gross sales by AtliQ Exclusive
08.	Best quarter by sold quantity in FY2020
09.	Channel contribution in FY2021
10.	Top 3 products per division in FY2021

Request No: 1

Provide the list of markets in which customer "AtliQ Exclusive" operates its business in the APAC region.

Result Grid	
	market
▶	Australia
	Bangladesh
	India
	India
	Indonesia
	Japan
	Newzealand
	Philiphines
	South Korea



Key Insights:

- AtliQ Exclusive is present in 8 APAC countries: Australia, Bangladesh, India, Indonesia, Japan, New Zealand, Philippines, and South Korea.
- India is the largest home market, while countries like Bangladesh and Philippines represent emerging growth opportunities with lower competitive saturation

Request No: 2

What is the percentage of unique product increase in 2021 vs. 2020?

(The final output contains these fields, unique_products_2020, unique_products_2021 percentage_chg)



Key Insights:

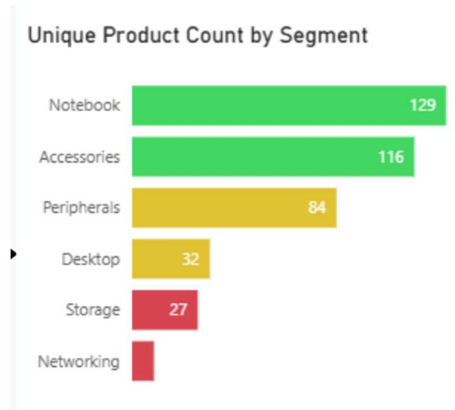
- Unique products grew from 245 in FY2020 to 334 in FY2021 a 36.33% increase year over year.
- AtliQ significantly expanded its product portfolio in FY2021, adding 89 new unique products. This aggressive expansion signals a deliberate strategy to capture broader market demand post-COVID-19 and compete across more categories.

Request No: 3

Provide a report with all the unique product counts for each segment and sort them in descending order of product counts.

(The final output contains 2 fields, segment, product_count)

Result Grid			Filter Rows:
	segment	product_count	
▶	Notebook	129	
	Accessories	116	
	Peripherals	84	
	Desktop	32	
	Storage	27	
	Networking	9	



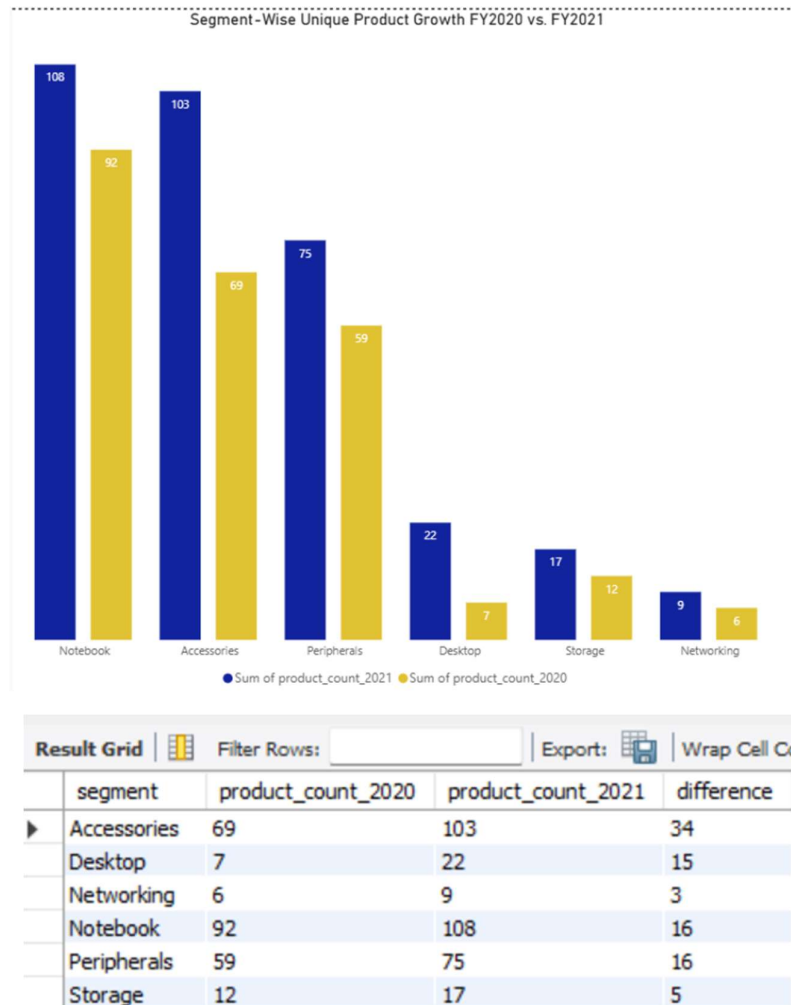
Key Insights:

- The product portfolio is heavily concentrated in Notebook and Accessories, which together account for nearly 62% of all unique products. Networking and Storage are significantly under-represented, with Networking having just 9 unique products a potential market gap
- Unique product is still a niche for the segments Network and Storage presenting a clear opportunity

Request No: 4

Follow-up: Which segment had the most increase in unique products in 2021 vs 2020?

(The final output contains these fields, segment, product_count_2020, product_count_2021, difference)



Key Insights:

- Accessories grew the most, adding 34 new unique products (69 → 103). Desktop had the highest percentage growth: from 7 to 22 products.
- This suggests a strong strategic push into the desktop market, possibly driven by the work-from-home demand surge that followed the pandemic

Request No: 5

Get the products that have the highest and lowest manufacturing costs/

(The final output should contain these fields, product_code, product manufacturing_cost)



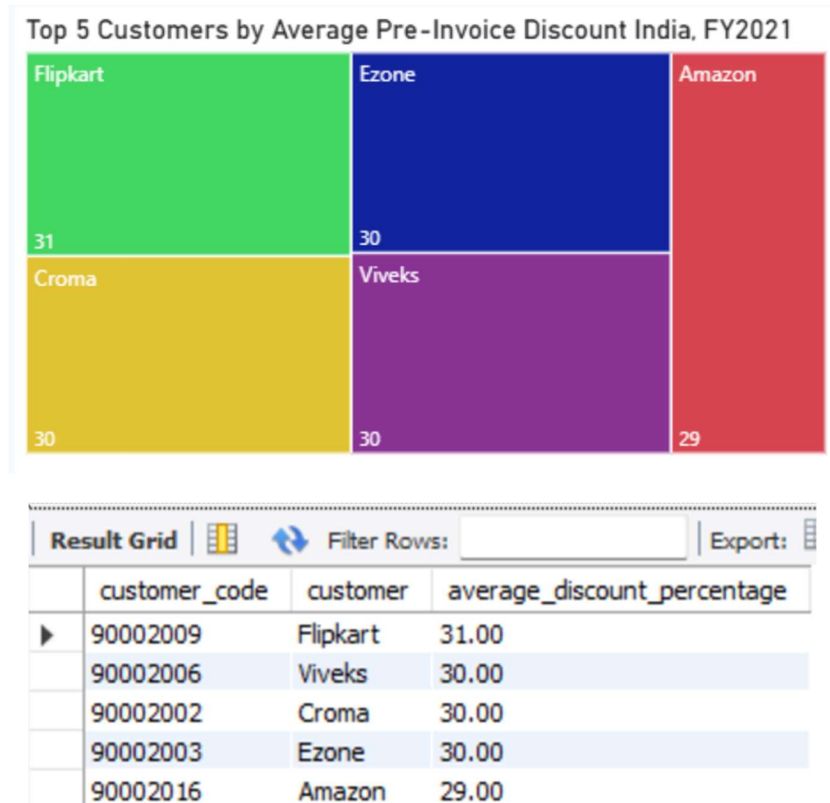
Result Grid Filter Rows: Export: Wrap Cell C				
	product_code	product	segment	manufacturing_cost
▶	A6120110206	AQ HOME Allin1 Gen 2	Desktop	240.5364
	A2118150101	AQ Master wired x1 Ms	Accessories	0.8920

Key Insights:

- AQ Home Allin1 Gen2 product from Desktop segment has the highest price as 240.54 USD the product is premium all-in-one desktop computers involves complex component assembly
- AQ Master Wired X1 Ms product from Accessories segment has the lowest price as 0.89 USD the product is Mouse and the \$0.89 peripheral likely benefits from mass production economies

Request No: 6

Generate a report which contains the top 5 customers who received an average high `pre_invoice_discount_pct` for the fiscal year 2021 and in the Indian market, (The final output contains these fields, `customer_code`, `customer average_discount_percentage`)



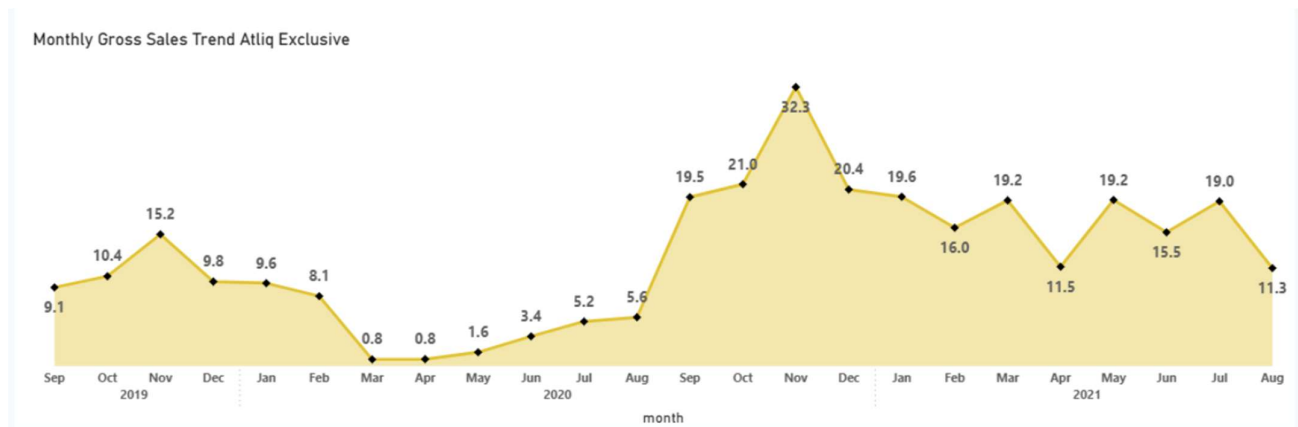
Key Insights:

- Flipkart led at 31%. Viveks, Croma, and Ezone all received 30%. Amazon received 29%. Discounts are tightly clustered among the top 5.
- Top 5 customers suggests a relatively standardised pre-invoice discount policy. Flipkart's marginal lead at 31% may reflect its dominant market share

Request No: 7

Get the complete report of the Gross sales amount for the customer “AtliQ Exclusive” for each month. This analysis helps to get an idea of low and high-performing months and take strategic decisions.

(The final report contains these columns: Month, Year, Gross sales Amount)



Key Insights:

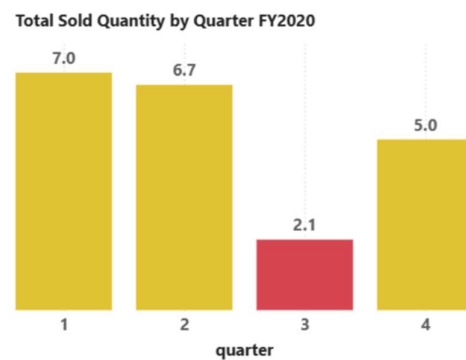
- Sales collapsed to \$0.77M in March 2020 (COVID lockdown). November 2020 was the peak at \$32.25M. Recovery was strong through FY2021 with consistent monthly sales of \$11M–\$19M.
- November 2020's peak at \$32.25M likely reflects Diwali-season buying and restocking by retailers ahead of year-end.

Request No: 8

In which quarter of 2020, got the maximum total_sold_quantity

(The final output contains these fields sorted by the total_sold_quantity, Quarter total_sold_quantity)

Result Grid			Filter Rows:
	quarter	total_sold_quantity	
▶	1	7.01	
	2	6.65	
	3	2.08	
	4	5.04	



Key Insights:

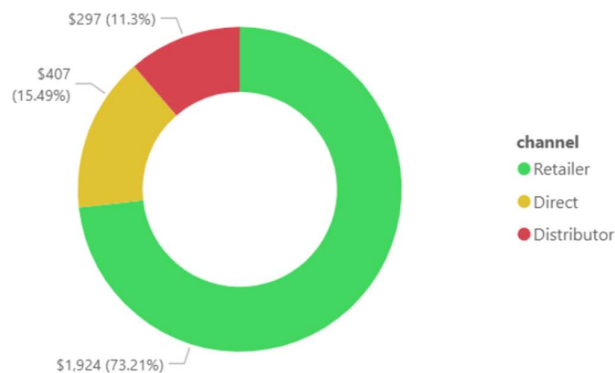
- Q1 (Sep–Nov 2019) had the highest sold quantity at 7.01M units. Q3 (Mar–May 2020) had the lowest at 2.08M units a direct impact of COVID-19.
- FY2020 Q1's strong performance reflects normal pre-pandemic business. The Q3 plunge to 2.08M is entirely attributable to COVID-19 lockdowns, which began in March 2020. The Q4 recovery to 5.04M units shows the market was already bouncing back by June - August 2020 driven by the demand

Request No: 9

Which channel helped to bring more gross sales in the fiscal year 2021 and the percentage of contribution?

(The final output contains these fields, channel, gross_sales_mln, percentage)

Gross Sales Contribution by Channel FY2021



Result Grid			
		Filter Rows:	
	channel	gross_sales_mln	percentage
▶	Retailer	1924.17	73.22
	Direct	406.69	15.48
	Distributor	297.18	11.31

Key Insights:

- Retailer channel dominated with \$1,924M (73.22%). Direct contributed \$407M (15.48%). Distributor contributed \$297M (11.31%). Amazon alone contributed 13.46% of total gross sales.
- The Retailer channel is overwhelmingly the largest revenue driver, with Amazon standing out as the single most impactful customer at 13.46% of all gross sales. The Direct channel at 15.48% reflects AtliQ's own-store and e-store operations (AtliQ e Store, AtliQ Exclusive), which together contributed over \$220M. Distributor revenue, while smallest, covers important markets where direct retail is not viable

Request No: 10

Get the Top 3 products in each division that have a high total_sold_quantity in the fiscal_year 2021

(The final output contains these fields, division, product_code, product, total_sold_quantity, rank_order)



division	product_code	product	total_sold_quantity	rank_order
N & S	A6720160103	AQ Pen Drive 2 IN 1	701373	1
N & S	A6818160202	AQ Pen Drive DRC	688003	2
N & S	A6819160203	AQ Pen Drive DRC	676245	3
P & A	A2319150302	AQ Gamers Ms	428498	1
P & A	A2520150501	AQ Maxima Ms	419865	2
P & A	A2520150504	AQ Maxima Ms	419471	3
PC	A4218110202	AQ Digit	17434	1
PC	A4319110306	AQ Velocity	17280	2
PC	A4218110208	AQ Digit	17275	3

Key Insights:

- N&S: AQ Pen Drive 2 IN 1 (701K units). P&A: AQ Gamers Ms (428K units). PC: AQ Digit (17.4K units). Volume varies dramatically across divisions N&S outsells PC
- The N&S division dominates on volume, with pen drives selling in the hundreds of thousands consistent with the institutional and bulk-purchase dynamics of low-cost storage products. P&A peripherals like gaming mouse also show strong consumer demand. PC volumes are far lower, reflecting the higher price point and longer replacement cycles of laptops and desktops.